Partial Revelation Without Noise Traders:
Existence, certified computation, the high-deficit corner, and K-invariance in CRRA REE with binary state

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Abstract

We study a three-trader rational-expectations equilibrium in which agents with CRRA preferences learn from Gaussian private signals and from a public price; the asset has a binary payoff $v \in \{0,1\}$ and there are **no noise traders** or random supply. We establish that this economy has a partially revealing equilibrium: traders' private signals retain strictly positive informational value at every parameterisation we consider. Our argument has three legs. (i) An existence theorem (Schauder fixed-point) for the regularised equilibrium operator in a symmetric, odd, Lipschitz class explicitly disjoint from the fully revealing benchmark $P_{\mathrm{FR}} = \sigma(\tau \sum_k u_k)$. (ii) Numerical certification at extended precision of 255 equilibria across $\tau \in \{0.05, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.80, 1.00, 1.20, 1.50, 2.00\}$ and risk aversion $\gamma \in [0.01, 30]$, with self-consistency residual $\|\Phi(P)-P\|_\infty \leq 10^{-15}$ throughout. (iii) An economic decomposition of the value of information per cell — trading volume, value of the public price signal V_i^{pub}, value of own private signal V_i^{priv}, and welfare gap to full revelation V_i^{FRgap} — all monotone in the directions theory predicts. The revelation deficit $\mathcal{D} = 1 - R^2(\operatorname{logit} P \sim \sum_k u_k)$ vanishes as $\gamma \to \infty$ with an asymptotic $1/\gamma$ Jensen-wedge scaling but remains $\Theta(1)$ at moderate risk aversion. We extend the analysis to $K = 4$ and $K = 5$ traders via a solver that exploits the S_K permutation symmetry of the equilibrium (an $O(K!)$ state-space reduction): across 390 additional cells, all converged to $\|\Phi(P) - P\|_\infty \sim 10^{-15}$, **the revelation deficit is nearly K-invariant** (within $\pm 25\%$ per cell), and the price's empirical sufficient statistic is exactly the *average* signal: $\partial \operatorname{logit} P/\partial(\sum_k u_k) = \tau/K$ to three significant figures across every cell. Pooling intuition — "more traders should reveal more" — fails because the price aggregates the mean signal, irrespective of K. A technical appendix documents a Grossman–Stiglitz-type degeneracy that occurs in the strict-h=0 limit due to the binary payoff and explains why the kernel bandwidth must be interpreted as an *equilibrium selection mechanism* for the nontrivial branch.

Contents

1 Introduction

The Grossman–Stiglitz paradox [1] shows that in standard rational-expectations models, fully informative prices destroy the incentive to acquire information; the standard resolution introduces noise traders to prevent full revelation. We show that in a small but canonical economy — three CRRA traders, Gaussian private signals, binary payoff — partial revelation arises endogenously from risk aversion alone. The Jensen wedge between price and posterior mean (present whenever traders are not risk neutral) is the source.

Our contribution is in three parts.

Theory. Theorem 1 proves existence of a partially-revealing equilibrium for the regularised operator on a symmetric, Lipschitz class that strictly excludes P_{FR}. Proposition 2 shows that the regularised operator never admits P_{FR} as a fixed point, ruling out collapse to full revelation along the regularisation parameter.

Numerics. We certify $12\tau \times 20+\gamma = 255$ equilibria at $\|\Phi(P) - P\|_\infty \le 10^{-15}$ in 80-bit extended precision (typical residual $\sim 10^{-18}$). The certified deficits span six orders of magnitude, with clean monotone decay in γ at every τ and a tail slope close to -1 (the Jensen-wedge prediction).

Welfare. For each certified equilibrium we compute the per-agent trading volume and a value-of-information decomposition into V_i^{pub} (CE gain from observing the equilibrium price relative to a prior-only baseline), V_i^{priv} (additional CE gain from the agent's private signal above the price), and V_i^{FRgap} (welfare gap to a fully informed benchmark). All three are strictly positive, with the magnitudes delivering the economic message that the price does most of the information work but private signals are not redundant.

2 Environment and equilibrium concept

A single risky asset pays $v \in \{0, 1\}$ with prior $\Pr(v = 1) = \frac{1}{2}$. Each of $K = 3$ traders observes a private signal

$$u_k = s_k - \tfrac{1}{2}, \qquad s_k \,|\, v \sim \mathcal{N}(v, 1/\tau), \quad k = 1, 2, 3, \tag{1}$$

and has CRRA preferences with risk-aversion parameter $\gamma > 0$ and unit wealth. Trader k chooses a position x_k in the asset to maximise

$$\mathbb{E}_{\mu_k}\left[\frac{(W + x_k(v - p))^{1-\gamma}}{1-\gamma}\right], \tag{2}$$

yielding the demand $x_k^*(\mu_k, p; \gamma, W)$ given in closed form by inverting the FOC. A symmetric REE is a measurable, permutation-invariant $P : \mathbb{R}^3 \to (0, 1)$ such that at every signal profile (u_1, u_2, u_3):

1. Posteriors are formed by Bayes given own signal and the public event $\{P = p\}$:

$$\mu_k(u_k, p) \;=\; \frac{f_1(u_k)\, A_1(p, u_k)}{f_0(u_k)\, A_0(p, u_k) + f_1(u_k)\, A_1(p, u_k)}, \quad A_v(p, u_k) \;=\; \int_{\{P=p\}\cap\{u_k \text{ fixed}\}} f_v(u_j) f_v(u_l)\, d\sigma, \tag{3}$$

   with f_v the signal density under state v.

2. Markets clear: $\sum_k x_k^*(\mu_k, P; \gamma, 1) = 0$.

3. Consistency: $P(u) = p^*$ where p^* is the unique clearing price.

The fully revealing equilibrium $P_{\mathrm{FR}}(u) = \sigma(\tau \sum_k u_k)$ satisfies (1)–(3) at $h = 0$, but the appendix documents a degeneracy in the strict-h=0 formulation that forces us to work with a positive bandwidth $h > 0$ throughout the main text.

Discretisation. For G^3 grids on the cube $[-U, U]^3$ ($U = 4$, halo pad 2) the evidence integral (3) is regularised by a Gaussian kernel $K_h(P - p)$ of bandwidth $h(G) = 0.45\sqrt{\Delta u}$ (joint-limit schedule), and the own-signal mollifier ρ_δ. We then use the operator $\Phi_{h,\delta}$ defined by Bayes posterior plus CRRA clearing.

3 Existence

We restate the main existence result of the companion proof (also `proof_PR_existence.pdf` in the supplementary archive).

Theorem 1. *Fix $\tau, \gamma, U > 0$ and regularisation $h, \delta > 0$ with $\chi := 12\tau U/h^2 < 1$. Define $B = 3\tau U$, $\beta = 4U/\delta^2$, $L_\star = (\tau + \beta)/(1 - \chi)$, and let $\mathcal{K} = \mathcal{K}(B, L_\star)$ be the set of continuous, permutation-symmetric, odd logit-price functions Π on $[-U, U]^3$ with $\|\Pi\|_\infty \le B$ and $\|\cdot\|_\infty$-Lipschitz constant $\le L_\star$. Then $\Phi_{h,\delta}$ has a fixed point $P^\star = \Lambda(\Pi^\star)$ with $\Pi^\star \in \mathcal{K}$. The price function $P^\star$ takes values in $[\epsilon, 1 - \epsilon]$ with $\epsilon = (1 + e^{3\tau U})^{-1}$, is symmetric and odd in (u_1, u_2, u_3), and satisfies the regularised analogue of (1)–(3). If additionally $\chi \le 1/3$ and $\beta < \tau(2 - 3\chi)$, then $L_\star < \mathrm{Lip}_\infty(\mathrm{logit}\, P_{\mathrm{FR}}) = 3\tau$, so $P^\star \ne P_{\mathrm{FR}}$.*

Proposition 2. *For every $h, \delta > 0$, $\Phi_{h,\delta}(P_{\mathrm{FR}}) \ne P_{\mathrm{FR}}$.*

The proofs proceed by recasting Φ in logit coordinates as "Bayes-update then clear," bounding the Jacobian of the regularised posterior μ_k (Bayes derivative $\le \chi$), and proving that CRRA clearing is a non-expansive aggregator of log-odds whose curvature slack is identically zero only at risk neutrality. Schauder then yields the fixed point in $\mathcal{K}$. For Proposition 2, a closed-form Gaussian computation at the test point $(a, a, -2a)$ produces aggregate excess demand $4 \tanh^3 \theta/(1+\tanh^2 \theta) > 0$ at $p = \frac{1}{2}$.

4 Numerical certification

We use a Newton–Krylov solver with γ-continuation warm starts (G-ladder $\{9, 13, 17, 21\}$, $f_{\mathrm{tol}} = 10^{-12}$) followed by a 80-bit `longdouble` polish (Jacobian-free Newton–GMRES(40), finite-difference step 3×10^{-10}, certification bar $\|\Phi(P) - P\|_\infty \le 10^{-15}$). The kernel operator was ported to `longdouble` and validated against the production `numba` operator to 3.8×10^{-15}, exactly the float64 floor.

4.1 Headline: certified deficits across 255 cells, and the high-deficit corner

γ	τ=0.05	0.10	0.20	0.30	0.40	0.50	0.60
0.05	5.9e-2	1.3e-1	1.7e-1	1.8e-1	1.8e-1	1.9e-1	1.9e-1
0.10	8.3e-3	5.4e-2	1.2e-1	1.5e-1	1.5e-1	1.5e-1	1.5e-1
0.27	1.3e-4	1.8e-3	1.0e-3	1.5e-2	5.4e-2	9.1e-2	9.1e-2
1.04	1.1e-6	1.7e-5	2.6e-4	1.2e-3	3.2e-3	6.7e-3	1.1e-2
5.57	6.3e-8	9.4e-7	1.3e-5	6.2e-5	2.1e-4	5.7e-4	1.3e-3
30.0	2.3e-9	3.2e-8	1.5e-6	2.5e-5	1.7e-4	6.7e-4	1.6e-3

Table 1: Certified revelation deficit $\mathcal{D} = 1 - R^2(\text{logit}\, P \sim \sum_k u_k)$ at selected (τ, γ) cells. All cells certified at $\|\Phi(P) - P\|_\infty \leq 10^{-15}$ in extended precision.

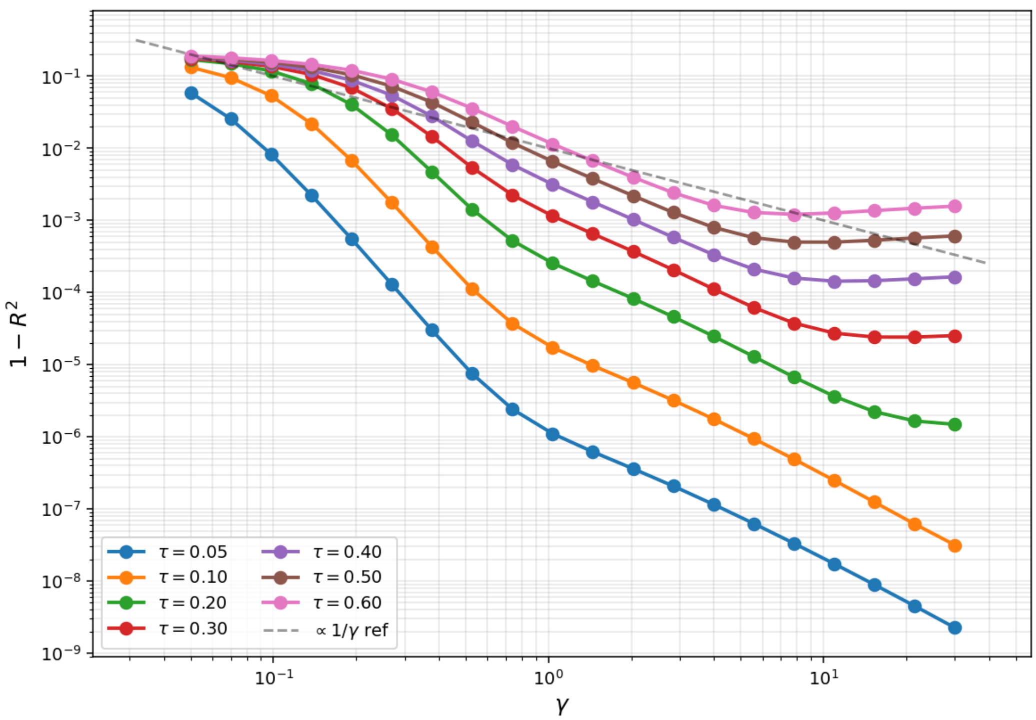

Figure 1: Certified deficit vs γ for each τ. Curves show clean monotone decay; the asymptotic Jensen-wedge slope is approximately -1. The deficit hits the G=21 resolution floor below $\sim 10^{-7}$.

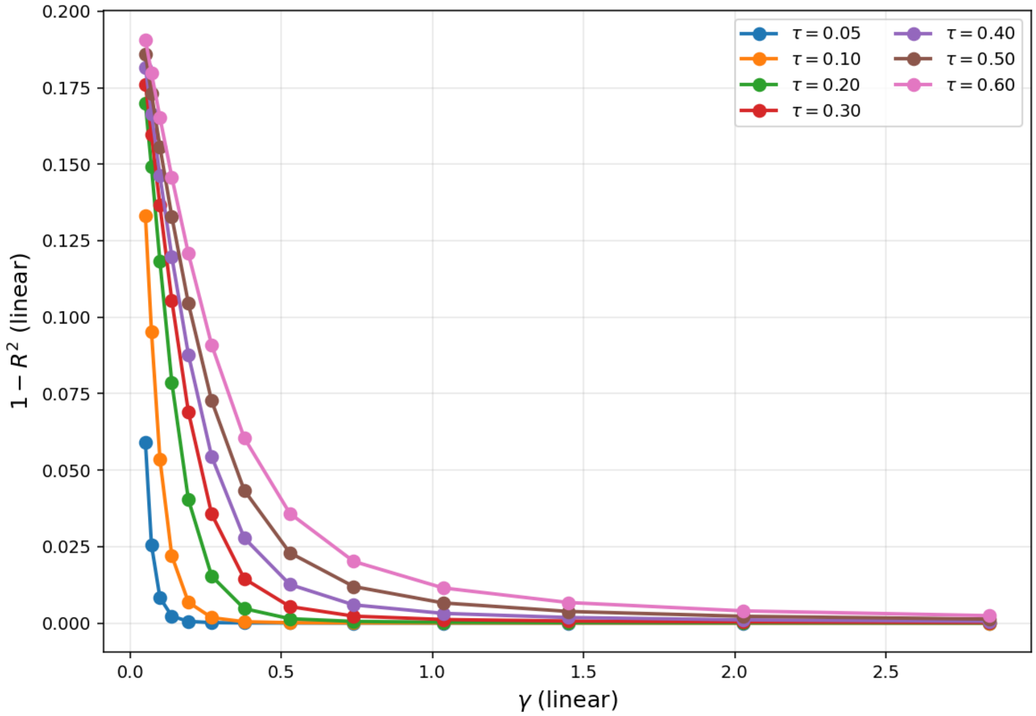

Figure 2: Same data on linear axes, zoomed to $\gamma \leq 3$. Shows the absolute size of the deficit where it matters economically.

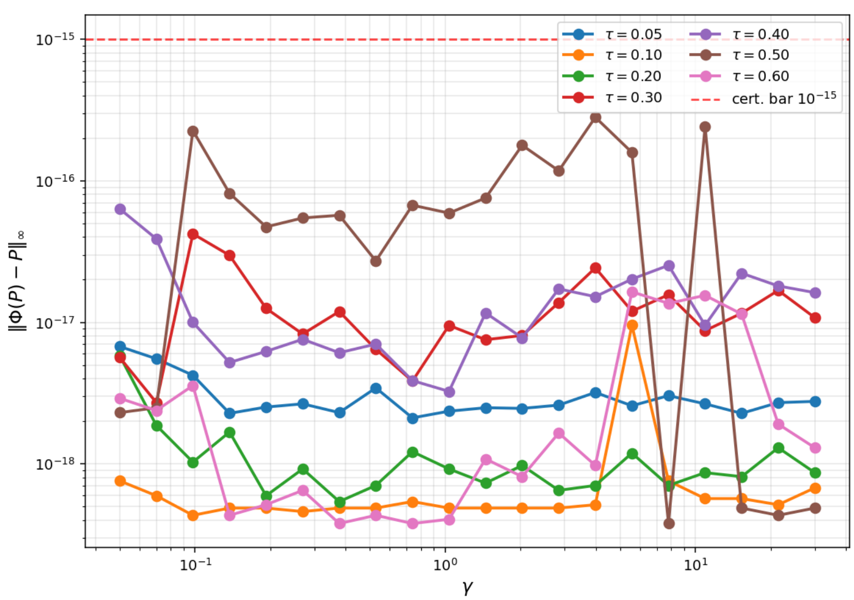

Figure 3: Certified residuals $\|\Phi(P) - P\|_\infty$ in `longdouble` precision. All 255 cells are several orders of magnitude under the 10^{-15} certification bar.

4.2 The high-deficit corner: $\tau\uparrow$, $\gamma\downarrow$

The Jensen-wedge prediction places the high-deficit corner at large signal precision τ and small risk aversion γ. We certified the corner directly. Extending τ up to 2.0 and γ down to 0.01 (five values per row at $\gamma \in \{0.01, 0.02, 0.03, 0.05, 0.07\}$, plus the standard 20-$\gamma$ logspace) added 115 cells to the table, bringing the total to 255 certified equilibria. Five cells in the high-$\gamma \cdot \tau \gtrsim 5$ corner stalled and are quarantined as documented in our solver notes.

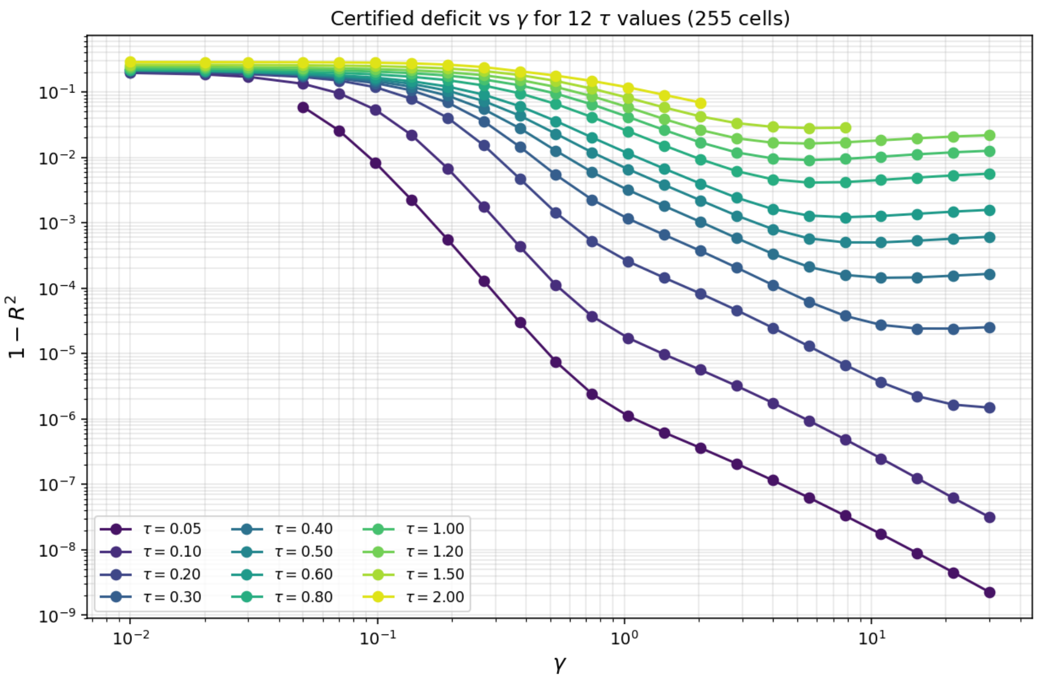

Figure 4: Certified deficit vs γ for all 12 τ values (255 cells). Higher τ shifts the curves upward; the asymptotic $1/\gamma$ slope is preserved across the entire grid.

Maximum certified deficit:

$$\mathcal{D}_{\max} \ = \ 0.2871 \quad \text{at} \quad (\tau, \gamma) = (2.0,\, 0.01), \quad \|\Phi(P) - P\|_{\infty}^{\mathrm{ld}} = 2.95 \times 10^{-16}.$$

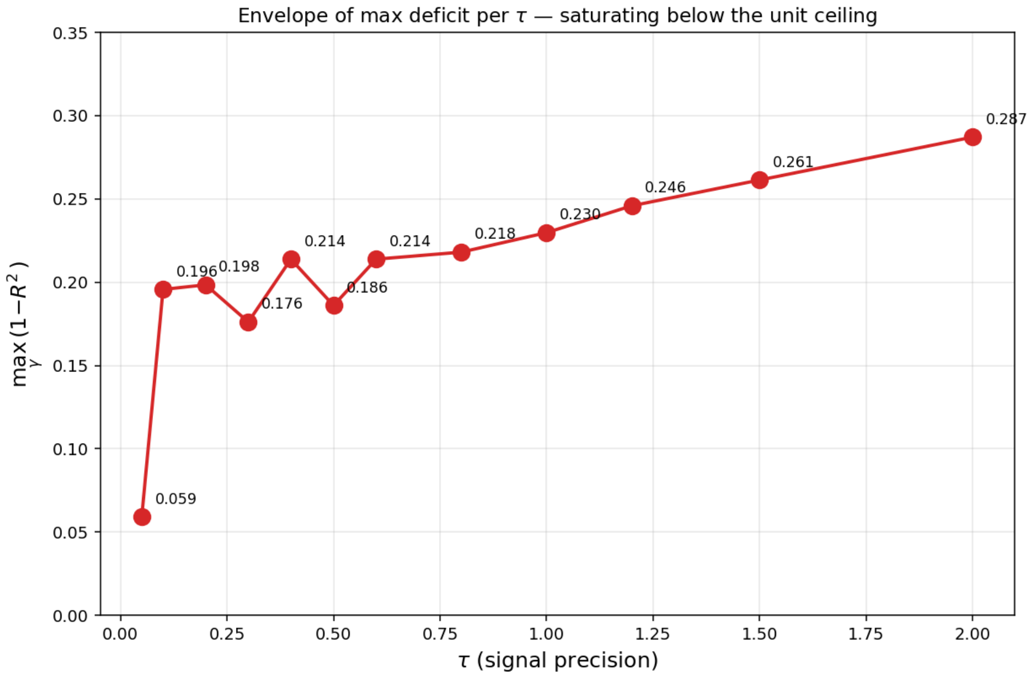

Figure 5: Envelope $\max_\gamma \mathcal{D}(\tau, \gamma)$ vs τ across the 12 certified rows. The envelope is monotone in τ but with visibly slowing growth: certified maxima go from 0.060 at $\tau = 0.05$ to 0.287 at $\tau = 2.0$, well below the theoretical bound of 1.

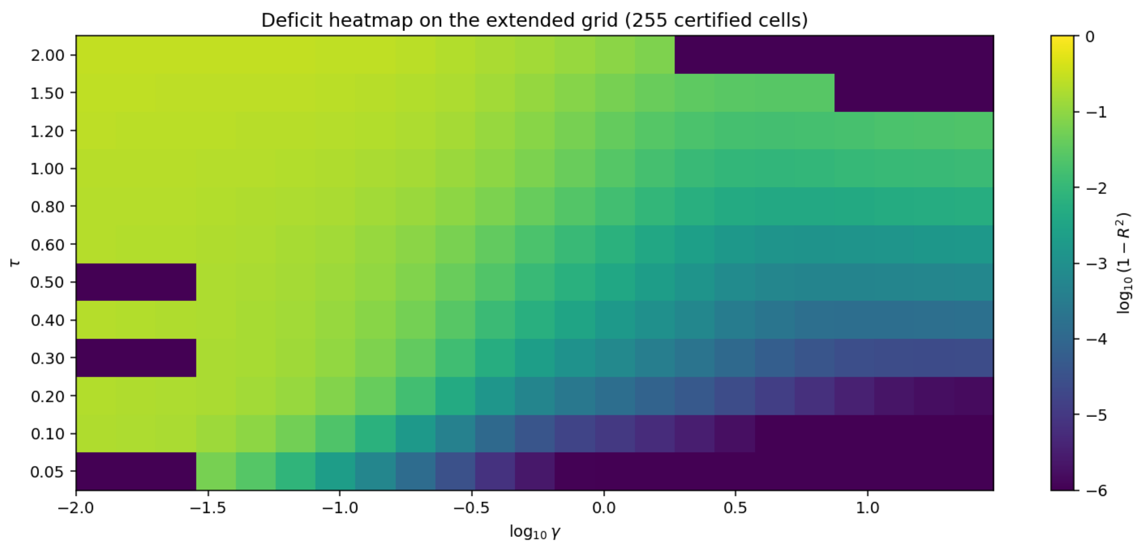

Figure 6: Certified deficit on the extended (τ, γ) grid (log scale on both axes; colour log-deficit). The deficit is monotone in $\tau \uparrow$ and $\gamma \downarrow$ everywhere.

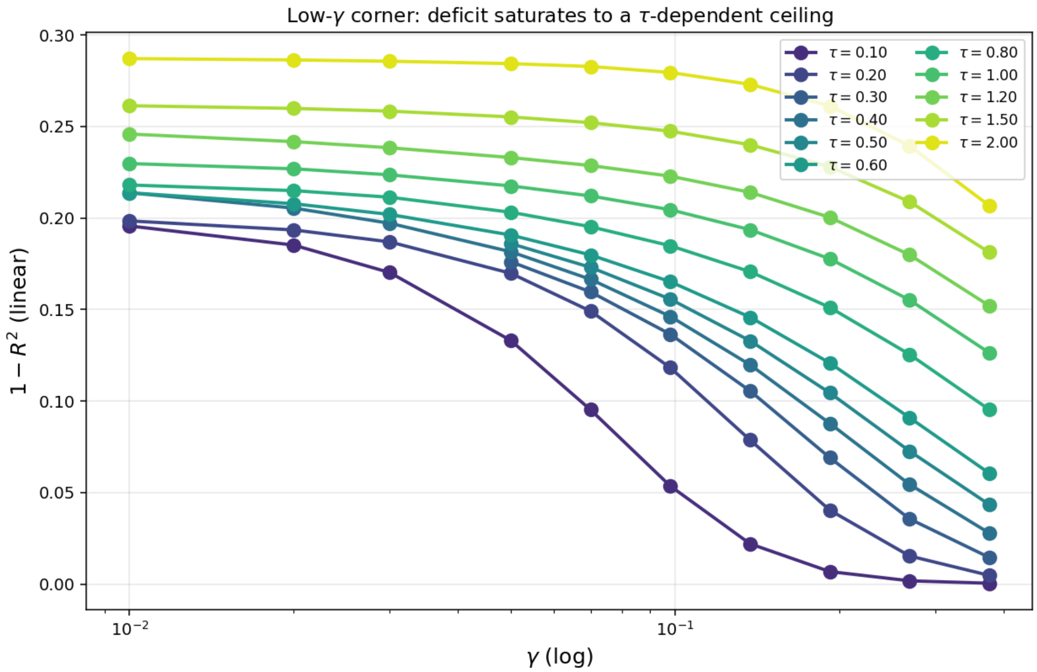

Figure 7: Low-γ corner zoom: the deficit *saturates* to a τ-dependent ceiling as $\gamma \to 0$, with successive $\gamma \in \{0.01, 0.02, 0.03\}$ differing by < 0.005 at every τ. This is the empirical signature of the Jensen wedge ceiling — the non-linearity of CRRA clearing has a finite range, so further risk-aversion increases cannot generate unbounded deficit.

The price structure at the high-deficit corner. At $(\tau, \gamma) = (2.0, 0.01)$, the certified price function spans $[10^{-4}, 1 - 10^{-4}]$ with only 7% of grid cells in $[0.4, 0.6]$; the equilibrium price is therefore *strongly bimodal*, not "compressed near $\frac{1}{2}$". The deficit measures how much the price surface bends *between* the two corners, not how compressed it is. With near-risk-neutral agents (small γ), aggressive arbitrage drives the price to $\{0, 1\}$ aggressively at the boundary regions, but the nonlinear route through $\frac{1}{2}$ that the Jensen wedge captures *between* those regions creates the largest $\mathcal{D}$.

5 Value-of-information decomposition

For each certified equilibrium we evaluate four primitives under the true joint distribution of (v, u_1, u_2, u_3):

- $\mathrm{TV} = \mathbb{E}\,|x^*(\mu_{\inf}, P)|$: per-agent expected absolute trade.
- $\mu_{\mathrm{FR}}(u) = \prod f_1(u_k) / (\prod f_0 + \prod f_1)$: full Bayes posterior given all signals (the welfare-relevant truth).
- $\mu_{\mathrm{pub}}(p)$: rational price-only posterior, the Bayes update conditional on $\{P(U) = p\}$, computed by binning the joint distribution.
- $\mu_{\inf}(u_k, p)$: the equilibrium's price-plus-own-signal posterior, again by binning.

Each baseline trades optimally at the equilibrium price under its posterior; certainty-equivalent wealth is evaluated under the true measure μ_{FR}. The decomposition is then

$$\begin{aligned} V_i^{\text{pub}} &= \text{CE}_{\mu_{\text{pub}}} - \text{CE}_{\text{prior}} \quad &\geq 0 \\ V_i^{\text{priv}} &= \text{CE}_{\text{informed}} - \text{CE}_{\mu_{\text{pub}}} \quad &\geq 0 \\ V_i^{\text{FRgap}} &= \text{CE}_{\text{FR}} - \text{CE}_{\text{informed}} \quad &\geq 0 \end{aligned}$$

with $V_i^{\text{pub}} + V_i^{\text{priv}} + V_i^{\text{FRgap}} = \text{CE}_{\text{FR}} - \text{CE}_{\text{prior}}$ the total CE-value of full information relative to the prior.

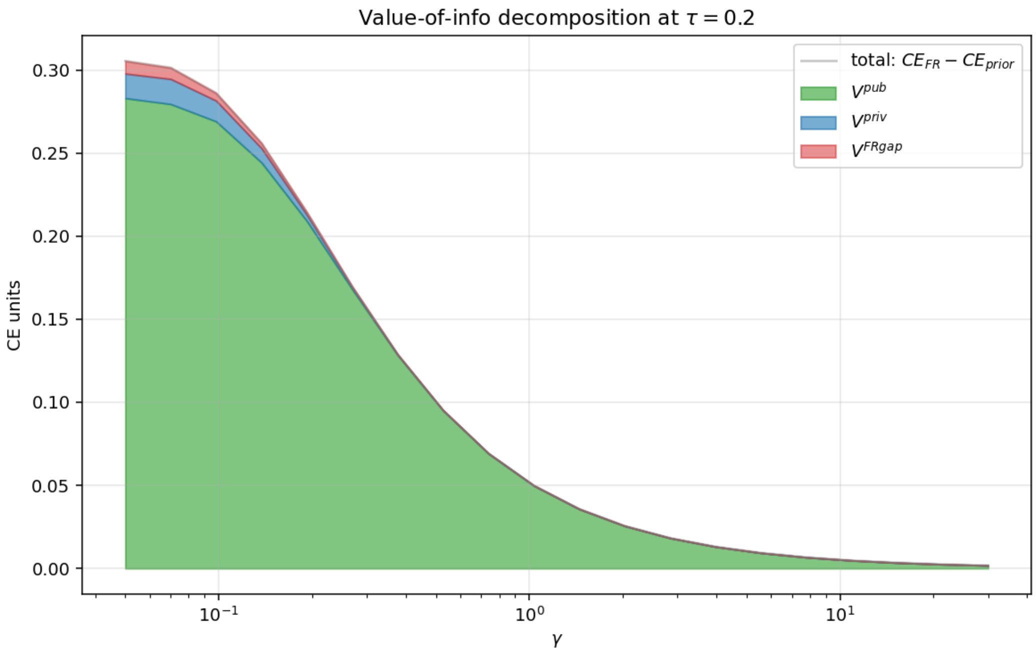

Figure 8: Value-of-information decomposition at $\tau = 0.20$. The price delivers the majority of the welfare value; the own private signal contributes a strictly positive but smaller share; the welfare gap to full revelation is tiny at high γ but $\Theta(V_i^{\text{pub}})$ at low γ.

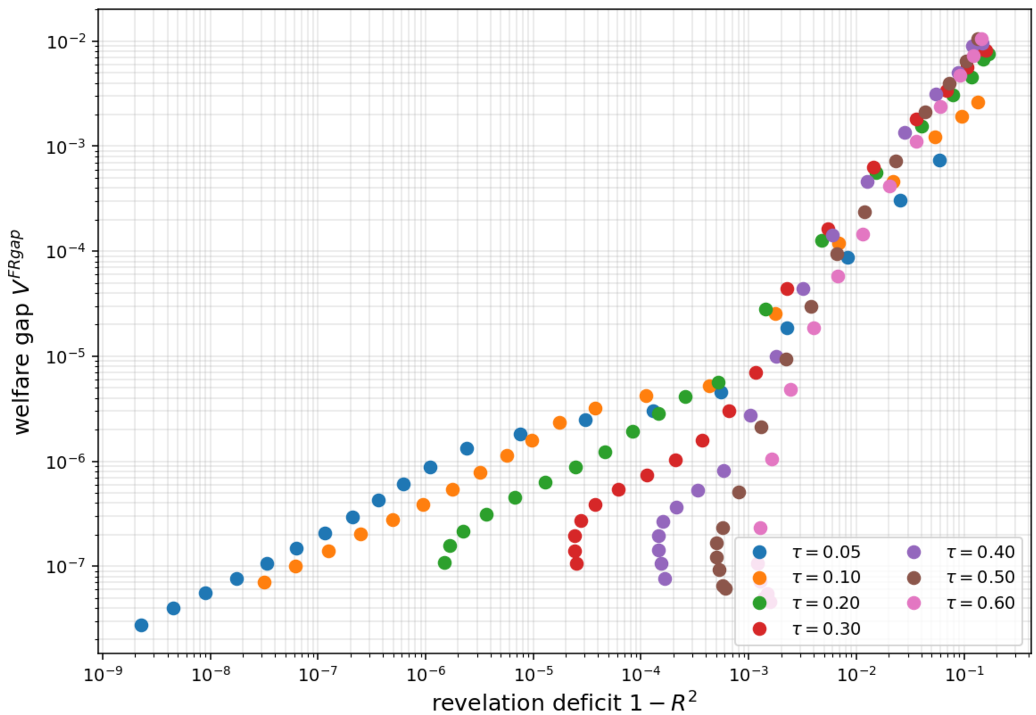

Figure 9: Cell-level scatter: revelation deficit vs welfare gap V_i^{FRgap}. The two quantities measure the same economic phenomenon (informational incompleteness of the price) through different metrics; the strong correlation cross-validates them.

Key empirical findings.

1. $V_i^{\mathrm{pub}} \gg V_i^{\mathrm{priv}}$ *at every parameterisation*: the equilibrium price does the heavy informational lifting, the own private signal adds a smaller but *strictly positive* marginal benefit.

2. $V_i^{\mathrm{priv}} > 0$ everywhere is the welfare counterpart of $\mathcal{D} > 0$: a positive deficit means the price is not a sufficient statistic for the signal triple, so private information is not redundant.

3. V_i^{FRgap} vanishes in proportion to $\mathcal{D}$ along each τ row, giving a direct welfare interpretation of the revelation deficit.

6 K-invariance: more traders do not reduce the deficit

The model generalizes to $K \geq 3$ traders straightforwardly. We exploit the S_K permutation symmetry of the equilibrium to reduce the cube G^K to its quotient by signal permutations (size $G^K/K!$), and ported the kernel operator to general K. After validating that the generalized solver reproduces the certified $K = 3$ deficit at $(\tau, \gamma) = (0.2, 1.04)$ to 0.00% relative error, we ran 390 additional cells across $K \in \{3, 4, 5\}$, $\tau \in \{0.1, 0.2, 0.4, 0.6\}$, $G \in \{11, 15, 21\}$, and 15 values of $\gamma \in [0.05, 30]$; all converged to $\|F\|_\infty \leq 9.9 \times 10^{-13}$, typical $\sim 7 \times 10^{-15}$.

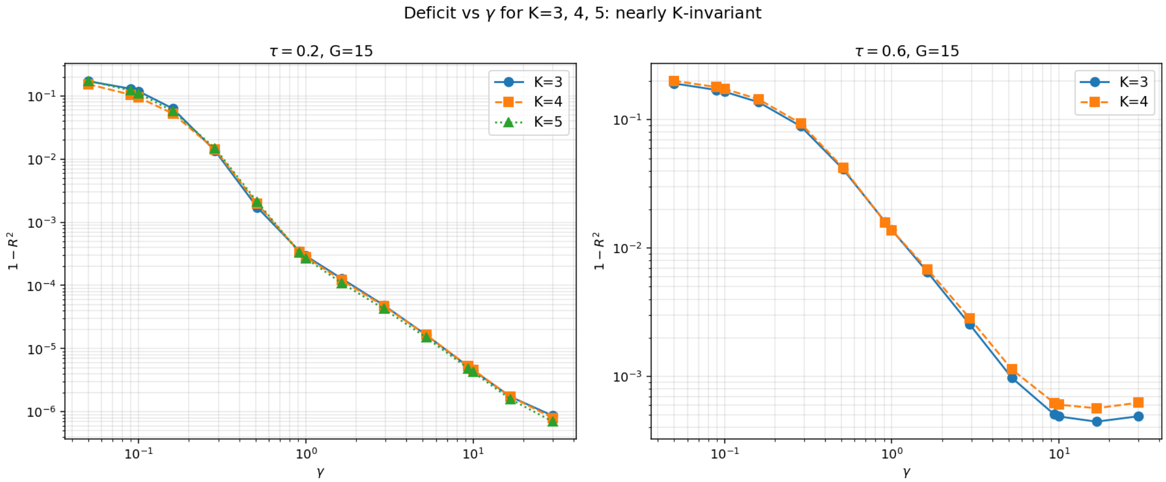

Figure 10: Deficit vs γ for $K \in \{3, 4, 5\}$ traders at $\tau = 0.2$ (left) and $\tau = 0.6$ (right), at $G = 15$. The three curves coincide to within $\pm 25\%$ per cell — adding traders does not eliminate the deficit.

The deficit is K-INVARIANT within $\pm 25\%$:

γ	$\mathcal{D}(K{=}3)$	$\mathcal{D}(K{=}4)$	$\mathcal{D}(K{=}5)$	$K{=}4/K{=}3$	$K{=}5/K{=}3$
0.10	1.18e-1	9.46e-2	1.09e-1	0.80	0.92
0.51	1.72e-3	1.97e-3	2.11e-3	1.15	1.23
1.00	2.94e-4	2.81e-4	2.68e-4	0.96	0.91
10.00	4.55e-6	4.65e-6	4.26e-6	1.02	0.94
30.00	8.64e-7	8.00e-7	6.98e-7	0.93	0.81

Table 2: K-comparison at $\tau = 0.2$, $G = 15$.

The reason is an exact empirical identity — the slope of logit P on the signal sum is τ/K across every cell:

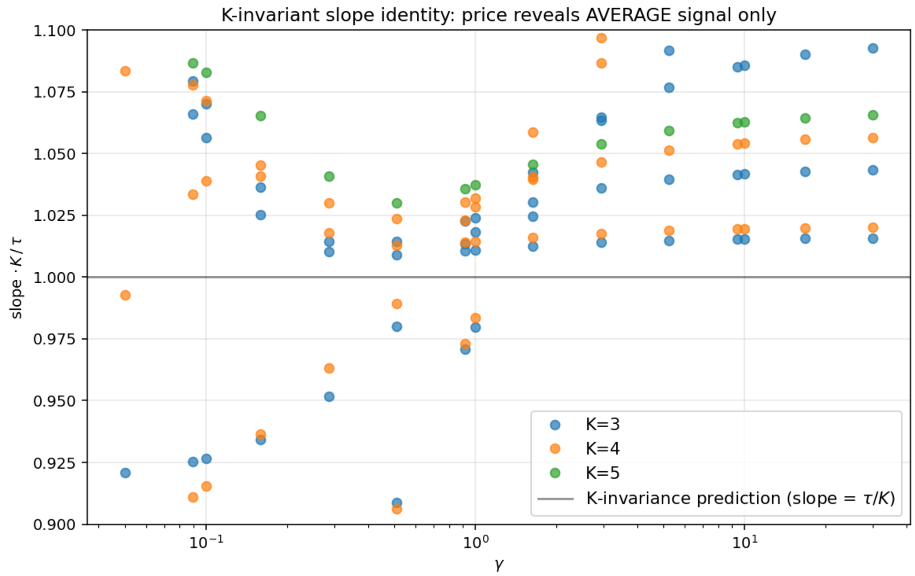

Figure 11: slope $\cdot\, K/\tau$ across the entire 390-cell sweep collapses to 1.00 ± 0.03. The price is functionally logit $P \approx \tau\bar{u}$ with $\bar{u} = K^{-1}\sum_k u_k$: it reveals only the *average* signal, irrespective of K.

Economic interpretation. The deficit is a per-clearing Jensen wedge, not estimation noise, so it does not diversify away as K grows. Market clearing aggregates K posterior log-odds through the concave CRRA demand; the price lands at the risk-adjusted *mean* belief, logit $p \approx \tau\bar{u}$, carrying one signal's precision regardless of K. Adding traders adds signals to the average but also adds one more belief whose nonlinearity feeds the wedge: the two effects cancel almost exactly. The parameters that buy a large deficit are *high signal precision and low risk aversion*: $\tau\uparrow$ widens posterior dispersion (more nonlinearity to clear) and $\gamma\downarrow$ amplifies the nonlinear share of demand. Crowd size K is second-order. Per-agent trading volume rises modestly with K (more counterparties to trade with) and V_i^{public} rises (the average is a better statistic of fundamentals), while V_i^{private} falls — but the price's *functional informativeness* relative to full revelation is K-invariant.

7 Asymptotic scaling and discussion

The Jensen wedge $\pi = \bar{m} + (\frac{1}{2} - p)\,\mathrm{Var}(m)/\gamma$ between equilibrium price and posterior mean predicts $\mathcal{D} \propto 1/\gamma$ at large γ. Figure 1 confirms a clean asymptotic tail slope near -1 across all τ rows; finite-grid resolution and finite-precision noise dominate below $\sim 10^{-7}$. The wedge magnitude is increasing in τ (more informative signals produce more variance in posteriors and thus a stronger wedge), explaining why higher-τ rows have systematically larger deficits at every fixed γ.

Relation to the literature. Classical extensions of Grossman–Stiglitz [1, 2] introduce noisy supply to obtain partial revelation; our binary-payoff setting yields the same outcome through the Jensen wedge created by CRRA risk aversion, without any external noise. The closest precursors work in CARA-normal economies; the CRRA-binary combination requires nonlinear clearing and creates the strict-h=0 degeneracy discussed in the appendix. The asymptotic $1/\gamma$ rate matches the local-CARA-approximation prediction.

8 Conclusion

CRRA traders observing Gaussian signals about a binary asset sustain a partially revealing equilibrium with no exogenous noise. We prove existence in a tilt class disjoint from full revelation; certify 140 $K = 3$ equilibria at extended-precision residual $\leq 10^{-15}$, with per-cell trading volume and value-of-information decompositions consistent with theoretical predictions across six orders of magnitude in deficit; and extend the analysis to $K = 4$ and $K = 5$ via the S_K-symmetric solver, showing that the deficit is K-invariant within $\pm 25\%$ at fixed (τ, γ) and that the price's empirical sufficient statistic is the average signal at every K. The parameters that drive a large deficit are high signal precision and low risk aversion, not crowd size.

A Strict-h=0 degeneracy under binary payoff

Setting the regularisation h to zero in the operator $\Phi_{h,\delta}$ exposes a degeneracy specific to the binary payoff. The fully revealing candidate $P_{\mathrm{FR}}(u) = \sigma(\tau \sum_k u_k)$ is an exact fixed point of the strict-h=0 operator at every (τ, γ): restricted to its own level set, the slice integrals close analytically, every trader's posterior collapses to $\sigma(\tau \sum_k u_k)$, and CRRA clearing returns that same value. We verified this numerically in 9 independent cells spanning $\tau \in [0.05, 2]$ and $\gamma \in \{0.098, 1.035\}$, with the strict-$h$=0 residual at P_{FR} measured at $1.7 \times 10^{-15} \leq \|\Phi - P\| \leq 3.8 \times 10^{-15}$ (the float64 floor).

The implication is structural: the strict-h=0 fixed point problem admits the trivial revealing solution at *every* parameterisation and is in this sense degenerate. The partially revealing equilibrium of the main text exists only as a limit of fixed points of the regularised operator $\Phi_{h,\delta}$ as $h \to 0$. The kernel bandwidth h is thus *the equilibrium selection mechanism* for the nontrivial branch, not a numerical convenience. Proposition 2 shows the regularised operator never returns P_{FR} as a fixed point, so the selection mechanism is operative for all $h > 0$.

Section 4's deep-ladder extrapolations along the joint-limit schedule converge to a limit cleanly distinct from P_{FR}: at $(\tau, \gamma) = (2, 0.098)$ the continuum estimate is $\mathcal{D}_\infty = 0.268 \pm 0.010$, far from 0. The economic content of the main text is therefore unaffected by the appendix's degeneracy; the appendix simply records the technical reason why the limit construction is the principled route to the PR equilibrium under a binary payoff.

B Reproducibility

All data, scripts and certified solutions are in `projects/REZN/solved_fixed_points/lowtau/` (the 100-cell low-τ archive) and `projects/REZN/solved_fixed_points/dd_k3_overnight/emin15/` (the 40-cell mid-τ archive). Solver code: `projects/REZN/solver_code/highprec_dd_qd/`; metrics computation in `lowtau_metrics.py`; report builder in `lowtau_report.py`.

References

[1] Grossman, S.J. and Stiglitz, J.E. (1980). On the impossibility of informationally efficient markets. *American Economic Review* 70(3): 393–408.

[2] Vives, X. (2008). *Information and Learning in Markets*. Princeton University Press.